Kaushal Jha

Ph.D. Researcher · ML & CFD · Porous Media

Golden, CO, USA
kaushal_jha@mines.edu · kaushal892jha@gmail.com
linkedin/kaushal-jha892 · github/WaRxChaMpioN

EDUCATION

Colorado School of Mines — *Jan 2026 – Present*
Ph.D., Hydrology | Advisor: Dr. Serveh Kamrava | Golden, CO, USA

- Research: ML-accelerated flow simulation in porous media for subsurface hydrology

Indian Institute of Technology Mandi — *Aug 2023 – Dec 2025*
M.Tech (Research), Computational Mechanics | Advisor: Dr. Pradeep Kumar | CGPA: 8.0/10

- Research: CFD, Radiative Heat Transfer, Porous Media, Monte Carlo methods

Dr. A.P.J. Abdul Kalam Technical University — *Jun 2023*
B.Tech, Mechanical Engineering | CGPA: 7.51/10 | Uttar Pradesh, India

- Thesis: Design and Development of a Portable Spot Welding Machine for Small-Scale Industries

RESEARCH & ACADEMIC PROJECTS

3D ML Surrogate for Lattice Boltzmann Flow in Porous Media — *Jan 2026 – Present*
Ph.D. Research, Colorado School of Mines | Advisor: Dr. Serveh Kamrava

- Reproducing Wang et al. (2021) using a 3D CNN surrogate to predict velocity fields from CT-scanned pore geometry.
- K4N8 architecture with continuity-based physics loss; $R^2 = 0.85$ at 64^3 resolution on Wendian HPC.
- Paper-quality 3D isosurface + velocity-weighted point-cloud visualisations.
- **TensorFlow** **Python** **HPC/SLURM** **LBM** **3D CNN** **VTK**

ML-Based Risk Classification for Oil & Gas Gathering Pipelines — *2025 – Present*
Collaborative Research | SPE paper in preparation [details to be updated upon completion]

- XGBoost pipeline for four-class risk classification (∼15k flowlines, 24:1 class imbalance).
- BorderlineSMOTE + TomekLinks; isotonic calibration; PR-AUC as primary metric.
- KDE (Kernal Density Estimation) Probability Distribution for Low/Medium/High/Severe labels.
- **XGBoost** **scikit-learn** **Python** **Monte Carlo** **Imbalanced-learn**

Tunnelling Through Fog for Long-Distance Laser Visibility — *Apr 2025 – Present*
M.Tech Research, IIT Mandi | Advisor: Dr. Pradeep Kumar

- Analysed complex refractive indices of ice and water across microwave, IR, and visible spectra.
- Laser–fog interaction simulation using Volume-of-Fluid and Monte Carlo ray-tracing in ANSYS Fluent.
- **MCRT** **ANSYS Fluent (VOF)** **Python** **Tecplot**

Radiation in Porous Media for Concentrated Solar Energy — *Apr 2024 – Dec 2025*
M.Tech Research, IIT Mandi | Advisor: Dr. Pradeep Kumar

- Coupled radiative-convective framework for a volumetric solar receiver using ANSYS Fluent + OpenFOAM.
- Reproduced pore-scale accuracy using a homogenised model, significantly reducing compute cost.
- **MCRT** **OpenFOAM** **ANSYS Fluent** **ICEM-CFD** **C/C++** **MATLAB**

Portable Resistance Spot Welding Machine — *Dec 2022 – May 2023*
B.Tech Major Project, AKTU | Mentor: Dr. Bishub Choudhary

- Low-cost (<$60) welder in Fusion 360 with 1500 A transformer; one-handed operable.
- Outcomes: 1 conference paper, 1 book chapter (Wiley 2024), 1 design patent filed (under review).
- **Fusion 360** **Product Design** **Resistance Welding** **Technical Writing**

WORK & INTERNSHIP EXPERIENCE

Aktiv Technologies Pvt. Ltd, Faridabad — *Jun – Jul 2023*
Intern, Design & Development

- Reverse-engineered components in SolidWorks; produced 3D models, 2D manufacturing drafts, product catalogues.
- **SolidWorks** **Reverse Engineering** **2D Drafting** **CorelDRAW**

Satyam Auto Components Pvt. Ltd, Gurgaon — *May – Jun 2023*

Graduate Engineer Trainee, Quality Assurance

- QA for resistance spot welding and stamping of fuel tanks and air filters on heavy press lines.
- **Quality Assurance** **Manufacturing Processes** **Heavy Press Line**

PUBLICATIONS

- **K. Jha**, S.S. Rathore, B. Mehta, P. Kumar. "Investigating the role of radiation in porous medium for concentrated solar energy applications." *Under review.*
- **K. Jha**, S.S. Rathore, B. Mehta, P. Kumar. "Estimation of Effective Thermal Conductivity of Porous Medium for Volumetric Solar Receiver Cavity-like System." *Submitted, IHMTC 2025.*
- **K. Jha**, S.K. Tamang, R. Kumar, B. Choudhary. "Enhancing Resistance Spot Welding Weld Quality." In *New Materials, Processing and Manufacturability*, Wiley, Ch. 5, Jul 2024. doi:10.1002/9781394212736.ch5
- **K. Jha**, A. Jumde, G. Kumar et al. "Design and Development of a Portable Resistance Spot Welding Machine." *Applied Mechanics and Materials*, Trans Tech, Feb 2024. doi:10.4028/p-xcpkt9

CONFERENCE PRESENTATIONS

- **K. Jha**. "Design and Development of Portable Resistance Spot Welding Machine." *AMMA 2023*, Amrita Vishwa Vidyapeetham, 2023.

HONORS, AWARDS & SCHOLARSHIPS

- **1st Place** — Poster Presentation, Symposium on Mechanical & Materials Engineering, IIT Mandi, 2024
- **GATE Qualified** — National-level engineering entrance exam, 2023
- **Research Assistantship (HTRA)** — Ministry of Education, Govt. of India, 2023
- **All India Rank 90** — NCAT (23,000+ participants), 2020

SKILLS

Simulation: OpenFOAM ANSYS Fluent ICEM-CFD SpaceClaim Tecplot

Programming: Python C/C++ MATLAB Fortran

ML / Data: TensorFlow scikit-learn XGBoost NumPy Pandas Matplotlib

HPC & Tools: SLURM/HPC Git LaTeX MS Office Ubuntu

CAD: SolidWorks Fusion 360

REFERENCES

- **Dr. Serveh Kamrava** — Assistant Professor, Colorado School of Mines *(Ph.D. Advisor)*
- **Dr. Pradeep Kumar** — Associate Professor, IIT Mandi *(M.Tech Advisor)* pradeepkumar@iitmandi.ac.in
- **Dr. Butunath Majhy** — Assistant Professor, IIT Guwahati butunath@iitg.ac.in
- **Dr. Bishub Choudhary** — Post-Doc, IISc Bangalore *(B.Tech Supervisor)* bishub.choudhary73@gmail.com